What drives sports car prices?

Analyzing how horsepower, torque, engine size & acceleration predict price across 1,007 sports cars from 38 brands

KEY HIGHLIGHTS

- Composite performance score is the strongest predictor (r = 0.60)
- HP vs Price r = 0.80 (excluding data-entry outlier)
- Price right-skewed — median $135K vs mean $383K

Background

Sports car prices vary widely across brands and performance specs. Consumers and dealers lack clear insight into which performance factors: horsepower, engine size, torque, or acceleration actually drive cost.

Dataset

Time-series	Production years 1990s–2020s
Records	1,007 rows × 14 columns
Features	HP, Torque, Engine Size, 0–60, Price

Analytical Model

Dependent: Price (USD)

Independent: HP, Torque, Engine Size, 0–60, Year

Methods: EDA → Correlation → Regression → Dashboard

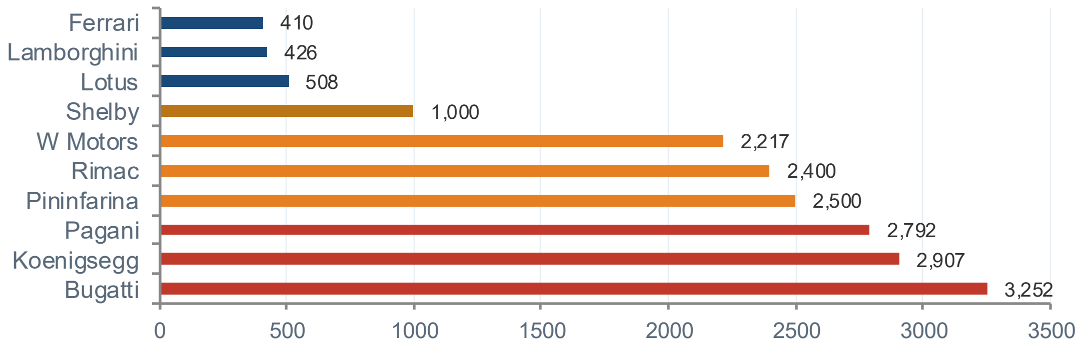
1,007
cars analyzed

38
unique brands

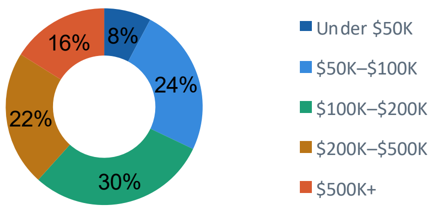
$383K
mean price

638 HP
avg horsepower

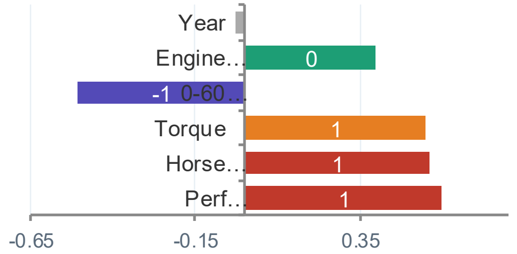
Average Price by Brand — Top 10 ($K)



Price Tier Distribution

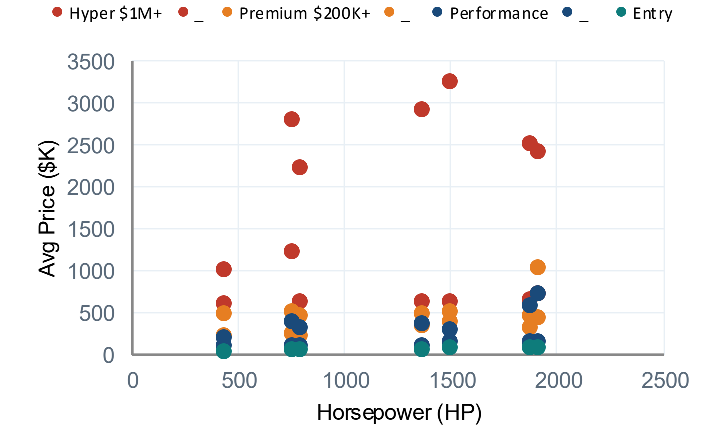


Correlation with Price (Pearson r)



Horsepower vs. Price

r = 0.80*



*r = 0.80 excluding one 10,000 HP data-entry error (Tesla)

Key Findings

- Performance score (composite of HP, torque, 0-60) is the strongest predictor of price at r = 0.60
- Horsepower r = 0.80 when excluding a 10,000 HP data-entry outlier — a key data quality finding
- Price is strongly right-skewed: mean $383K vs median $135K, driven by ultra-luxury hypercars
- Engine size is a moderate predictor (r = 0.40) — weakened by electric vehicles with no displacement
- Brand premium is real: Bugatti ($3.25M avg) commands 8× the dataset mean beyond performance alone